GSA of Model Output With Dependent Inputs

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Outline

Notations & Assumptions

- $y = \mathcal{M}(\boldsymbol{x})$ the scalar QoI
- $\boldsymbol{x} = (x_1, x_2, \ldots, x_d) \sim p_x = \dfrac{\partial^d F_x}{\partial x_1 \partial \ldots \partial x_d}$
- $x_i \sim p_i = \dfrac{\mathrm{d}F_i}{\mathrm{d}x_i}$, the marginal pdf ($F_i$ cdf) of x_i
- $p_{i|j} = \dfrac{\partial F_{i|j}}{\partial x_i}$, the conditional pdf (cdf) of x_i over x_j, ...
- $\mathcal{D} = \{0, 1, \ldots, d\}$, $\mathcal{D}_{-i} \bigcap \{i\} = \emptyset$, $\mathcal{D}_{+i} \bigcap \{i\} \neq \emptyset$
- $\boldsymbol{x} = (\boldsymbol{x}_\alpha, \boldsymbol{x}_{-\alpha})$ with $\boldsymbol{x}_\alpha \bigcap \boldsymbol{x}_{-\alpha} = \emptyset$
- $\alpha = \{i_1, \ldots, i_k\} \subseteq \mathcal{D} \Leftrightarrow \boldsymbol{x}_\alpha = \boldsymbol{x}_{i_1, \ldots, i_k} = (x_{i_1}, \ldots, x_{i_k})$

Marginal and conditional cdfs known

Rosenblatt Transformations

Assumption: Assumed known $F_{\alpha|\beta}$, $\forall \alpha \subseteq \mathcal{D}$ and $\alpha \cap \beta = \emptyset$

RT transforms $\boldsymbol{x} \sim p_x$ into $\boldsymbol{u} \sim \mathcal{U}(0,1)^d$

$$\begin{cases} u_{i_1} = F_{i_1}(x_{i_1}) \\ u_{i_2} = F_{i_2|i_1}(x_{i_2}|x_{i_1}) \\ \vdots \\ u_{i_d} = F_{i_d|i_1,\dots,i_d}(x_{i_d}|\boldsymbol{x}_{-i_d}) \end{cases} \tag{1}$$

Issue: RT is not unique. The conditional cdfs not always known.

Sobol' HDMR

RT turn $\boldsymbol{x} \sim p_x$ into $\boldsymbol{u} \sim \mathcal{U}(0,1)^d$ (i.e. $p_u = 1$). As a consequence it also turns $\mathcal{M}(\boldsymbol{x})$ into $f(\boldsymbol{u})$. Sobol' hdmr is as follows,

$$f(\boldsymbol{u}) = \sum_{\alpha \subseteq \mathcal{D}} f_\alpha(\boldsymbol{u}_\alpha) \tag{2}$$

with $f_0 = \mathbb{E}[f(\boldsymbol{u})]$, and, $\int_0^1 f_\alpha(\boldsymbol{u}_\alpha)\mathrm{d}u_k = 0, \forall k \in \alpha$

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As a result:

- **Sobol's hdmr is not unique as the RTs are not unique (unless $p_x = \prod_{i=1}^d p_i$)**
- **The f_α's are pairwise $\perp$**

How do we interpret the Sobol' indices so calculated?

Interpretation of the sensitivity indices: Mara & Tarantola (RESS, 2012)

The interpretation of the sensitivity indices of u_i as those of x_i is the following:

- Because $u_{i_1} = F_{i_1}(x_{i_1})$, the sensitivity indices of u_{i_1} are interpreted as **the full sensitivity indices of x_{i_1} that account for its mutual contribution due to its dependence on the other variables**
- Because $u_{i_2} = F_{i_2|i_1}(x_{i_2}|x_{i_1})$, the sensitivity indices of u_{i_2} are interpreted as those of x_{i_2} that account for its mutual contribution due to its dependence on the remainders **except** x_{i_2}

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- The sensitivity indices of u_{i_3} are those of x_{i_3} that account for its mutual contribution due to its dependence on the remainders **except** (x_{i_1}, x_{i_2})
- $\vdots$
- The sensitivity indices of u_{i_d} are those of x_{i_d} **without its mutual dependence contribution due to its dependence on the remainders**

Independent & Full SI's

Hence, one can define two types of indices: the Independent and the Full Sensitivity Indices
The first one measures the importance of a variable without the mutual contribution due to its dependence on the other variables, while the second one does.

N.B.: But because the Rosenblatt transformation is not unique, we need to consider several RTs to get the overall indices (i.e. full+independent).

Examples

Suppose that the RT is applied in the following order $(x_1, \ldots, x_d)$, that is

$$\begin{cases} u_1 = & F_{x_1}(x_1) \\ u_2 = & F_{x_2|x_1}(x_2|x_1) \\ \vdots & \\ u_d = & F_{x_d|\boldsymbol{x}_{\sim d}}(x_d|\boldsymbol{x}_{\sim d}) \end{cases}$$

Variance-based: (see Mara and Tarantola 2012)

$$S_1^{full} = \frac{\mathrm{Var}\left[\mathbb{E}\left[y|u_1\right]\right]}{\mathrm{Var}\left[y\right]} \tag{3}$$

$$ST_1^{full} = \frac{\mathbb{E}\left[\mathrm{Var}\left[y|\boldsymbol{u}_{\sim 1}\right]\right]}{\mathrm{Var}\left[y\right]} \tag{4}$$

$$S_d^{ind} = \frac{\mathrm{Var}\left[\mathbb{E}\left[y|u_d\right]\right]}{\mathrm{Var}\left[y\right]} \tag{5}$$

$$ST_d^{ind} = \frac{\mathbb{E}\left[\mathrm{Var}\left[y|\boldsymbol{u}_{\sim d}\right]\right]}{\mathrm{Var}\left[y\right]} \tag{6}$$

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pdf-based: (see Mara and Becker 2021)

$$\delta_1^{full} = \frac{1}{2}\int_{\mathbb{R}}\int_0^1 |p_y - p_{y|u_1}|\mathrm{d}y\mathrm{d}u_1 \qquad (3)$$

$$\delta_d^{ind} = \frac{1}{2}\int_{\mathbb{R}}\int_0^1 |p_y - p_{y|u_d}|\mathrm{d}y\mathrm{d}u_d \qquad (4)$$

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<u>cdf-based</u>:

$$\tau_1^{full} = \int_0^1 \sup |F_y - F_{y|u_1}| \mathrm{d}u_1 \quad (3)$$

$$\tau_d^{ind} = \int_0^1 \sup |F_y - F_{y|u_d}| \mathrm{d}u_d \quad (4)$$

Suppose that the RT is applied to $(x_2, x_3, \ldots, x_d, x_1)$, that is

$$\begin{cases} u_2 = & F_{x_1}(x_2) \\ u_3 = & F_{x_3|x_2}(x_3|x_2) \\ \vdots & \\ u_d = & F_{x_d|x_2,x_3,\ldots,x_{d-1}}(x_d|x_2, x_3, \ldots, x_{d-1}) \\ u_1 = & F_{x_1|\boldsymbol{x}_{\sim 1}}(x_1|\boldsymbol{x}_{\sim 1}) \end{cases}$$

Variance-based:

$$\begin{aligned} S_2^{full} &= \frac{\mathrm{Var}\left[\mathbb{E}\left[y|u_2\right]\right]}{\mathrm{Var}\left[y\right]} \\ ST_2^{full} &= \frac{\mathbb{E}\left[\mathrm{Var}\left[y|\boldsymbol{u}_{\sim 2}\right]\right]}{\mathrm{Var}\left[y\right]} \\ S_1^{ind} &= \frac{\mathrm{Var}\left[\mathbb{E}\left[y|u_1\right]\right]}{\mathrm{Var}\left[y\right]} \\ ST_1^{ind} &= \frac{\mathbb{E}\left[\mathrm{Var}\left[y|\boldsymbol{u}_{\sim 1}\right]\right]}{\mathrm{Var}\left[y\right]} \end{aligned}$$

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pdf-based:

$$\begin{aligned} \delta_2^{full} &= \frac{1}{2}\int_{\mathbb{R}}\int_0^1 |p_y - p_{y|u_2}|\mathrm{d}y\mathrm{d}u_2 \\ \delta_1^{ind} &= \frac{1}{2}\int_{\mathbb{R}}\int_0^1 |p_y - p_{y|u_1}|\mathrm{d}y\mathrm{d}u_1 \end{aligned}$$

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<u>cdf-based</u>:

$$\tau_2^{full} = \int_0^1 \sup |F_y - F_{y|u_2}| \mathrm{d}u_2 \tag{5}$$

$$\tau_1^{ind} = \int_0^1 \sup |F_y - F_{y|u_1}| \mathrm{d}u_1 \tag{6}$$

By considering all possible circular permutations (i.e., with $(x_3, x_4, \ldots, x_d, x_1, x_2)$, $(x_4, x_5, \ldots, x_d, x_1, x_2, x_3)$, etc.) one can compute the overall full and independent sensitivity indices of interest.

Computational cost: for given data methods like the BSPCE only N model runs are necessary, for the MC IA-estimator $(2 + 4d)N$ model runs are required.

Marginal and conditional cdfs unknown

If the u-values are unknown, the only sensitivity indices that can be computed are the **full sensitivity indices**, namely,

$$S_i^{full} = \frac{\mathrm{Var}\left[\mathbb{E}\left[y|x_i\right]\right]}{\mathrm{Var}\left[y\right]}$$

$$\delta_i^{full} = \frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}} |p_y - p_{y|x_i}| p_{x_i} \mathrm{d}y \mathrm{d}x_i$$

$$\tau_i^{full} = \int_{\mathbb{R}} \sup |F_y - F_{y|x_i}| p_{x_i} \mathrm{d}x_i$$

For the Sobol' indices we can find an estimate $\mathbb{E}\left[y|x_i\right]$ by regression. There is also the possibility to estimate the total-independent Sobol' indices

$$T_i^{ind} = \frac{\mathbb{E}\left[\mathrm{Var}\left[y|\boldsymbol{x}_{\sim i}\right]\right]}{\mathrm{Var}\left[y\right]} = 1 - \frac{\mathrm{Var}\left[\mathbb{E}\left[y|\boldsymbol{x}_{\sim i}\right]\right]}{\mathrm{Var}\left[y\right]}$$

for this we must fit a d-1 dimensional function to estimate $\mathbb{E}\left[y|\boldsymbol{x}_{\sim i}\right]$.

Exercise

Let us consider the ballistic model that computes the distance travelled by a rocket,

$$dist = \frac{v_0^2 \sin(2\theta)}{g}$$

The uncertain input variables, assumed **independent** of each other, are $\boldsymbol{x} = (v_0, \theta, g)$. Their pdf are given below,

v_o	Initial velocity (m/s)	$[0.5; 1.5]$
θ	Intial inclination (deg)	$[20; 45]$
g	Gravity acceleration (m/s^2)	$[9.5; 10]$

Estimate the variance-based sensitivity indices when (v_0, θ, g) are uniformly distributed over the lower part of the hyper-rectangle $[0.5, 0.15] \times [20, 45] \times [9.5, 10]$.

The RT transformation to use is:

$$\frac{x_1 - x_{1,min}}{x_{1,max} - x_{1,min}} = 1 - (1 - u_1)^{1/3},$$

$$\frac{x_2 - x_{2,min}}{x_{2,max} - x_{2,min}} = \left(1 - (1 - u_2)^{1/2}\right)(1 - u_1)^{1/3},$$

$$\frac{x_3 - x_{3,min}}{x_{3,max} - x_{3,min}} = (1 - (1 - u_3))\,(1 - u_2)^{1/2}\,(1 - u_1)^{1/3}$$

Shapley effect

The Shapley effect (Owen 2014) is a concept stemming from game theory (Shapley 1953). It is defined as follows

$$Sh_i = \sum_{\alpha \subseteq \mathcal{D}_{-i}} \frac{|\alpha|!\,(d-1-|\alpha|)!}{d!} \left(S^{clo}_{\alpha \cup i} - S^{clo}_{\alpha} \right)$$

where, $S^{clo}_{\alpha} = \frac{\mathrm{Var}[\mathbb{E}[f(\boldsymbol{x})|\boldsymbol{u}_{\alpha}]]}{\mathrm{Var}[f(\boldsymbol{x})]} = S^{full,clo}_{\alpha}$ is the first-order effect of the group of inputs $\boldsymbol{x}_{\alpha}$.

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Therefore, we need to compute the overall *full* sensitivity indices?

Main obstacle: We need to compute the overall $2^d - 1$ Sobol' indices.

Some References

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